NVMe Host Accelerator v1.0

PB058 (v1.0) February 19, 2019 **LogiCORE IP Product Brief**

Introduction

The Xilinx® NVMe Host Accelerator LogiCORE IP Product Guide core provides a simple and efficient interface to multiple NVMe drives, thereby offloading the CPU for the I/O queues and enabling a high throughput storage solution inside an FPGA. The core provides a path for either (or both) software or hardware module(s) to interface with it. Standard AXI memory map and streaming interfaces allow for easy integration. The NVMe Host Accelerator core is fully parameterizable and offers multiple customizations for a resource efficient implementation tailored according to your requirements. The Admin queue is managed by software (SW) and the IP offloads the following functions from the CPU:

- Submission Queue (SQ) doorbell management across multiple queues
- Completion Queue (CQ) doorbell management across multiple queues
- Building NVMe specification compliant submission queue command entry
- Completion queue entry parsing

Additional Documentation

A full product guide is available for this core. Access to this material can be requested by clicking on this registration link: https://www.xilinx.com/member/forms/registration/nvmeha.html.

Features

The features of the IP are categorized into Application Interface features and SSD Side Interface features.

Application Interface

- Support for AXI4 mapped slave to interface with software along with AXI Lite interface for register access
- Support for AXI4-Stream to interface with hardware design modules
- Support for configurable number of SQs per SSD (independent number of queues for hardware interface and software interface)
- SQs written into a FIFO like interface for both memory mapped and streaming ports
- Independently configurable depth of SQ depth on software and hardware interfaces
- Completion entries for memory mapped interface communicated using interrupt

- Completion entries for streaming interface communicated by generating streaming data

SSD Side Interface

- Support for configurable number of back-end SSDs
- Round-robin arbitration of all SQs to SSD
- AXI4 mapped master to ring SQ tail doorbells and CQ head doorbells
- AXI4 mapped slave to read from SQs and write to CQs

IP Facts

LogiCORE™ IP Facts Table	
Core Specifics	
Supported Device Family[1]	UltraScale+ and UltraScale
Supported User Interfaces	AXI4-Lite, AXI4-Stream, and AXI4-Memory mapped
Resources	Performance and Resource Use web page (registration required)
Provided with Core	
Design Files	Encrypted RTL
Example Design	Verilog
Test Bench	Not Provided
Constraints File	Xilinx Constraints File
Simulation Model	Not provided
Supported S/W Driver[2]	Standalone and Linux
Tested Design Flows[2]	
Design Entry	Vivado® Design Suite
Simulation	For supported simulators, see the Xilinx Design Tools: Release Notes Guide.
Synthesis	Not Provided
Support	
Provided by Xilinx at the Xilinx Support web page	

Notes:

1. For a complete list of supported devices, see the Vivado® IP catalog.
2. For the supported versions of the tools, see the Xilinx Design Tools: Release Notes Guide.

Overview

The NVMe Host Accelerator core manages the control path of multiple connected back end drives. The SSDs are responsible for pushing and pulling data from the respective buffers as provided in the SQ entry command and the data path doesn't pass through the IP. NVMe Host Accelerator has configurable software and hardware access mechanisms. The design can be configured to enable or disable either of the application interfaces (software or hardware). The number of back end NVMe drives and the number of SQs per drive, per interface are all configurable. The NVMe Host Accelerator assumes a 1:1 mapping between the SQs and the CQs.

Licensing and Ordering

This Xilinx® LogiCORE™ IP module is provided under the terms of the Xilinx Core License Agreement. The module is shipped as part of the Vivado® Design Suite. For full access to all core functionalities in simulation and in hardware, you must purchase a license for the core. To generate a full license, visit the product licensing web page. Evaluation licenses and hardware timeout licenses might be available for this core. Contact your local Xilinx sales representative for information about pricing and availability.

For more information about this core, visit the NVMe Host Accelerator product lounge.

License Checkers

If the IP requires a license key, the key must be verified. The Vivado® design tools have several license checkpoints for gating licensed IP through the flow. If the license check succeeds, the IP can continue generation. Otherwise, generation halts with an error. License checkpoints are enforced by the following tools:

- Vivado Synthesis
- Vivado Implementation
- write_bitstream (Tcl command)

Note: IP license level is ignored at checkpoints. The test confirms a valid license exists. It does not check IP license level.

Technical Support

Xilinx provides technical support on the Xilinx Community Forums for this LogiCORE™ IP product when used as described in the product documentation. Xilinx cannot guarantee timing, functionality, or support if you do any of the following:

- Implement the solution in devices that are not defined in the documentation.
- Customize the solution beyond that allowed in the product documentation.
- Change any section of the design labeled DO NOT MODIFY.

To ask questions, navigate to the Xilinx Community Forums.

Documentation Navigator and Design Hubs

Xilinx® Documentation Navigator (DocNav) provides access to Xilinx documents, videos, and support resources, which you can filter and search to find information. To open DocNav:

- From the Vivado® IDE, select **Help → Documentation and Tutorials**.
- On Windows, select **Start → All Programs → Xilinx Design Tools → DocNav**.
- At the Linux command prompt, enter `docnav`.

Design Hubs:

- In DocNav, click the **Design Hubs View** tab.
- On the Xilinx website, see the Design Hubs page.

Note: For more information on DocNav, see the Documentation Navigator page on the Xilinx website.

Revision History

The following table shows the revision history for this document.

Section	Revision Summary
02/19/2019 Version 1.0	
Initial release	N/A

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